



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Collected Falcon 9 launch data via the SpaceX REST API and web scraping (Wikipedia)
 - Performed data wrangling to engineer a binary Class label (1 = landing success, 0 = failure)
 - Conducted EDA using visualization (Matplotlib/Seaborn) and SQL queries
 - Built an interactive Folium map (site markers, 1000m radius circles, MarkerCluster color-coded by outcome) and a Plotly Dash dashboard (site dropdown, payload range slider, success-rate pie chart, payload-vs-success scatter plot)
 - Trained and tuned four classification models (Logistic Regression, SVM, Decision Tree, KNN) using GridSearchCV with 10-fold cross-validation

Introduction

- **GitHub Repository:** <https://github.com/buHtiGDasda/IBM> (contains all completed notebooks: data collection API, web scraping, data wrangling, EDA visualization, EDA SQL, Folium map, Plotly Dash app, and predictive analysis)

Introduction

- Project background and context
 - SpaceX advertises Falcon 9 launches at \$62M, far below competitors' ~\$165M, largely because it reuses the first stage instead of discarding it
 - Predicting whether the first stage will land successfully allows estimation of launch cost, useful for a company wanting to compete against SpaceX
- Problems
 - What factors determine whether the first stage successfully lands (payload mass, orbit type, launch site, flight number)?
 - Which launch sites and orbit types have the highest first-stage recovery rates?
 - Can a machine learning model reliably predict landing outcome before launch?

Section 1

Methodology

Methodology

Executive Summary

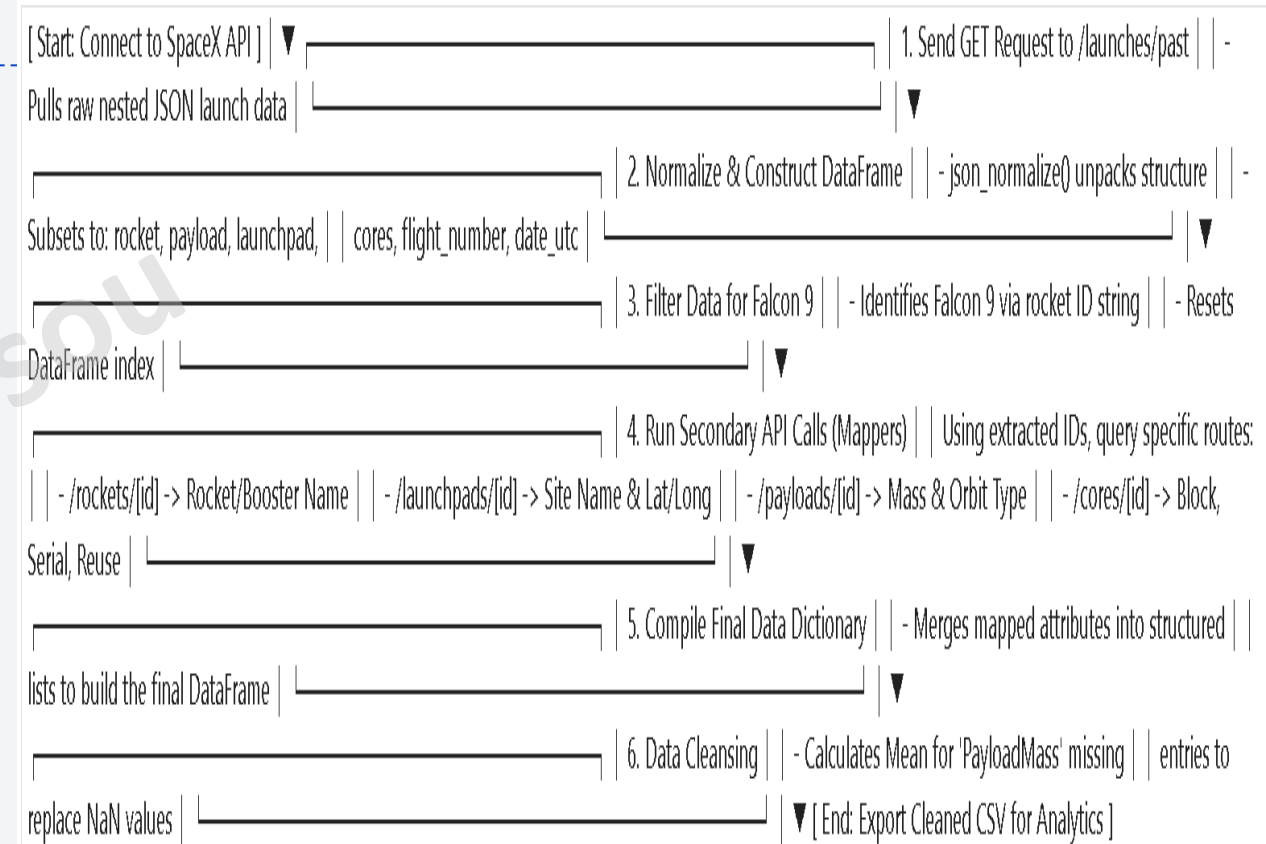
- Data collection methodology:
 - Collected via SpaceX REST API (GET requests, JSON → DataFrame) and web scraping of Wikipedia launch records with BeautifulSoup
- Perform data wrangling
 - Cleaned missing values, filtered to Falcon 9 only, and engineered a binary Class target from landing outcomes
- Perform exploratory data analysis (EDA) using visualization and SQL
 - Analyzed launch trends by flight number, payload mass, orbit type and launch site using Seaborn plots and SQL queries
- Built interactive maps with Folium (circles, DivIcon labels, MarkerCluster)
- Built an interactive dashboard with Plotly Dash (dropdown, range slider, pie chart, scatter plot)
- Perform predictive analysis using classification models
 - Trained Logistic Regression, SVM, Decision Tree, and KNN; tuned hyperparameters with GridSearchCV (10-fold CV) and compared test accuracy

Data Collection

- Two complementary sources were used to build a complete dataset of Falcon 9 launches
- **SpaceX API:** GET request to `/v4/launches/past` → JSON normalized into a DataFrame → filtered to Falcon 9 → enriched with booster, launch site, payload, and core data via helper functions
- **Web scraping:** Wikipedia Falcon 9 launch table parsed with BeautifulSoup, headers cleaned, landing outcomes normalized
- Both sources were merged and cleaned to produce the final structured dataset used for wrangling and modeling
- *API request → JSON → DataFrame → filter Falcon 9 → enrich fields ;*
- *Scraping → HTML parse → clean table → merge*

Data Collection – SpaceX API

- **Primary API Request:** A GET request is sent to the SpaceX API endpoint (`https://api.spacexdata.com/v4/launches/past`) to fetch historical launch data as a JSON object.
- **Data Flattening:** The JSON response is converted into a pandas DataFrame using `json_normalize()` to unpack nested structural data.
- **Filtering for Falcon 9:** The dataset is filtered using `.loc` to isolate rows where the rocket column equals the core ID for Falcon 9, discarding other vehicles like Falcon 1 or Falcon Heavy.
- **Custom API Helper Functions:** Loop functions (`getBoosterVersion`, `getLaunchSite`, `getCore`, `getPayload`) are executed to make secondary REST calls using IDs extracted from the primary launch data.
- **Feature Extraction:** Discovered IDs are replaced with human-readable names (e.g., coordinates, core serial numbers, payload mass, and landing outcomes).
- **Data Cleaning & Missing Value Handling:** Null values found in the payload mass column are calculated and replaced using the column's `.mean()` value to preserve dataset integrity.



- GitHub URL of the completed SpaceX API calls notebook: <https://github.com/buHtiGDasda/IBM/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Data Collection - Scraping

- **Target Snapshot Setup:** Define a hardcoded target snapshot URL (`static_url`) along with custom User-Agent string dictionaries to mirror browser traffic and handle server-side tracking safely.
- **HTML Parsing Instance:** Send a network request using the `requests` library and wrap the raw HTML payload body inside a `BeautifulSoup` instance to expose element tracking utilities.
- **Header Sanitization:** Strip formatting elements (`<br>`, `<a>`, `<sup>`) inside structural table rows to compile uniform string keys, ignoring empty entries or digit row counters.
- **Helper Normalization:** Execute modular string cleaning operations (like utilizing `unicodedata.normalize` on landing statuses and checking index substrings for unit labels like "kg") to extract raw data figures from complex table blocks cleanly.

```
[1. Environment & Setup]
    | (Install & import required packages: requests, beautifulsoup4, pandas)
    v
[2. Send HTTP Request]
    | (Send requests.get() to the target Wikipedia oldid URL with headers)
    v
[3. Parse HTML Document]
    | (Pass response.text to BeautifulSoup using 'html.parser')
    v
[4. Extract Table Columns]
    | (Isolate 'table' elements and pull names from table headers <th>)
    v
[5. Parse & Clean Rows] ← Iterate through table rows (<tr>) and text cells (<td>)
    | (Apply customized helper functions to scrub strings & handle formats)
    v
[6. Build DataFrame & Export]
    | (Append structured list of dicts into a Pandas DataFrame)
    v
[Target Dataset]
```

- GitHub URL of the completed web scraping notebook:
<https://github.com/buHtiGDasda/IBM/blob/main/jupyter-labs-webscraping.ipynb>

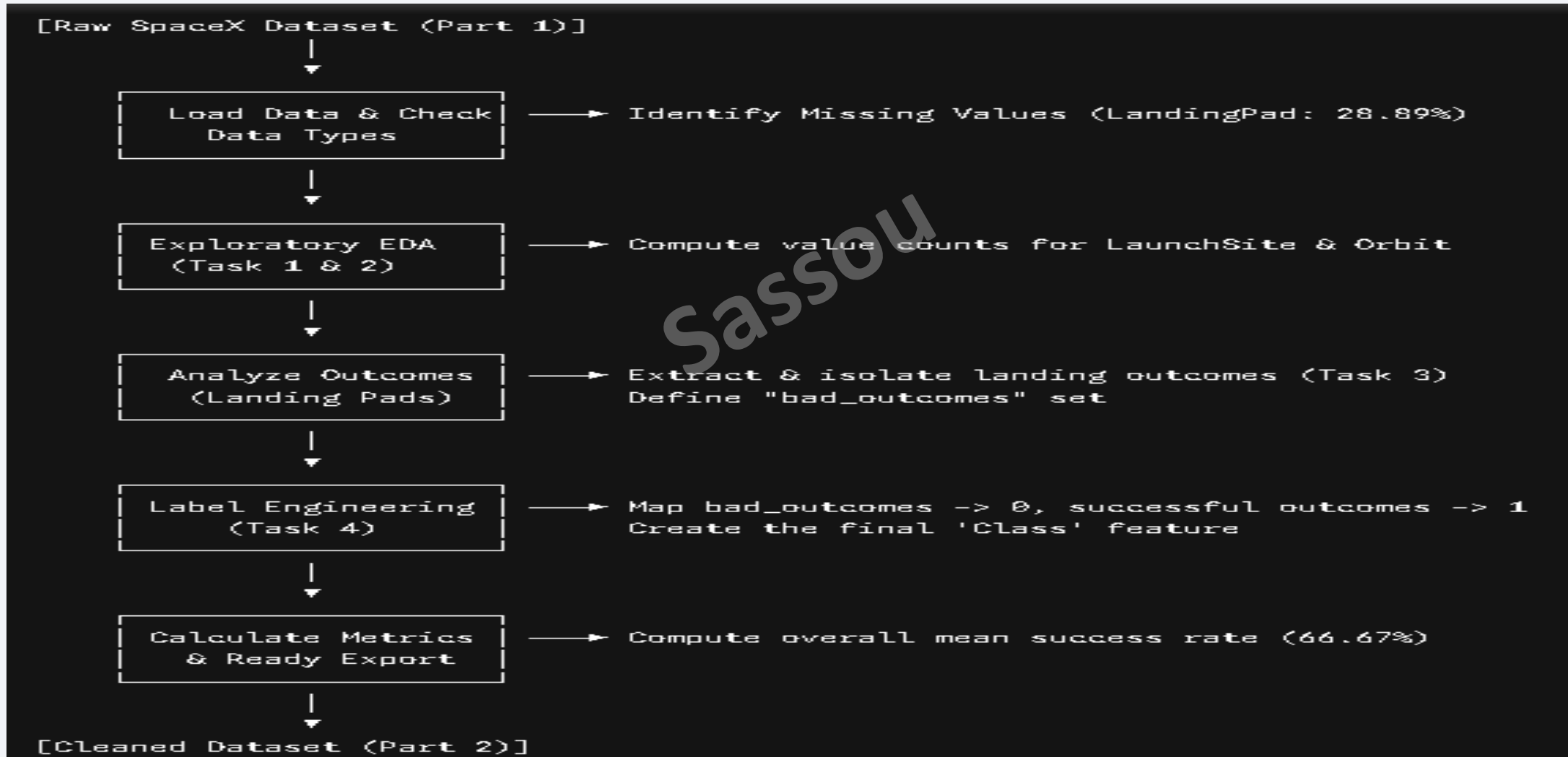
Data Wrangling

- The goal of this data wrangling phase was to clean, explore, and convert raw SpaceX launch outcomes into binary training labels (1 for a successful landing, 0 for an unsuccessful landing) suitable for supervised machine learning models.

Key Processing Steps

- **Libraries and Data Loading:** * Imported `pandas` and `numpy`.
 - Loaded the raw dataset (`dataset_part_1.csv`) containing 17 original features such as `FlightNumber`, `PayloadMass`, `Orbit`, `LaunchSite`, and `Outcome`.
- **Missing Value Analysis:** * Calculated the percentage of missing data per attribute. Only the `LandingPad` column contained missing values (approximately 28.89%).
- **Exploratory Feature Analysis:** * Computed launch frequencies across different launch sites (`LaunchSite`), noting that **CCAFS SLC 40** had the highest activity (55 launches).
 - Analyzed orbit destination frequencies (`Orbit`), revealing **GTO** (27) and **ISS** (21) as the most frequent.
- **Outcome Categorization:** * Extracted unique values from the `Outcome` column and filtered out unsuccessful landing outcomes into a `bad_outcomes` set: `{'False ASDS', 'False Ocean', 'False RTLS', 'None ASDS', 'None None'}`.
- **Target Label Creation:** * Engineered a new binary column called **Class** using a list comprehension: if an outcome fell within `bad_outcomes`, it was labeled 0, otherwise 1.
 - Determined the baseline success rate by calculating the mean of the `Class` column, which resulted in **66.67%**.
- GitHub URL: <https://github.com/buHtiGDasda/IBM/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Data Wrangling



EDA with Data Visualization

Chart Plotted

- **Type of Chart:** Categorical scatter plot (sns.catplot).
- **Variables Displayed:**
 - **X-axis:** FlightNumber (representing continuous launch attempts).
 - **Y-axis:** PayloadMass (Payload Mass in kg).
 - **Hue (Color overlay):** Class (representing the launch outcome—whether the first stage landed successfully or not).

Why This Chart Was Used

- **To Analyze Trends Over Time:** It allows to see if SpaceX's landing success rate improved as the number of flight attempts increased.
 - **To Observe the Impact of Weight:** It helps determine whether heavier or lighter payloads (PayloadMass) have a noticeable effect on the success of the first-stage return.
 - **To Identify Patterns at a Glance:** Overlaying the launch outcome with color makes it easy to visually grasp that as the flight number increases, the first stage is more likely to land successfully, even when carrying more massive payloads.
- GitHub URL of completed EDA with data visualization notebook: <https://github.com/buHtiGDasda/IBM/blob/main/edadataviz.ipynb>

EDA with SQL

- **Unique Launch Sites:** Retrieved the unique names of all launch sites used in the space missions (`SELECT DISTINCT "Launch_Site" ...`).
- **Filtered Launch Sites:** Displayed 5 mission records where the launch site name started with the string 'CCA' (`WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5`).
- **Total NASA Payload Mass:** Calculated the total sum of payload mass carried by boosters assigned to 'NASA (CRS)' (`SELECT SUM("PAYLOAD_MASS__KG_") ... WHERE "Customer" = 'NASA (CRS)'`).
- **Average Payload Mass:** Determined the average payload mass carried specifically by the booster version 'F9 v1.1' (`SELECT AVG("PAYLOAD_MASS__KG_") ... WHERE "Booster_Version" = 'F9 v1.1'`).
- **First Successful Ground Pad Landing:** Identified the earliest date on which a successful landing on a ground pad was achieved using the minimum date function (`SELECT MIN("Date") ... WHERE "Landing_Outcome" = 'Success (ground pad)'`).
- **Successful Drone Ship Boosters:** Listed distinct booster versions that successfully landed on a drone ship and carried a payload mass strictly between 4,000 kg and 6,000 kg.
- **Mission Outcome Counts:** Grouped and counted the total number of successful and failed mission outcomes (`SELECT "Mission_Outcome", COUNT(*) ... GROUP BY "Mission_Outcome"`).
- **Maximum Payload Boosters:** Used a subquery to find and list all booster versions that carried the maximum overall payload mass recorded in the dataset.
- **2015 Drone Ship Failures:** Extracted records from the year 2015 to display the converted month name, landing outcome, booster version, and launch site for missions that ended in a drone ship landing failure.
- **Ranked Landing Outcomes:** Grouped, counted, and ranked all landing outcomes in descending order based on their frequency for missions occurring between 2010-06-04 and 2017-03-20.

Build an Interactive Map with Folium

- **folium.Map:** The base map object itself, initially centered at the NASA Johnson Space Center coordinates with a specified zoom level, and later re-initialized to show the whole region.
- **folium.Circle:**
 - An initial test circle with a radius of 1000 meters centered around the NASA Johnson Space Center.
 - Multiple circles (each with a 1000-meter radius) added iteratively at the exact coordinate locations of the four distinct SpaceX launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E).
- **folium.map.Marker with DivIcon:** Text-label markers placed alongside the circles to explicitly display the names of the locations on the map surface (e.g., "NASA JSC" or the specific launch site names).
- **folium.Popup:** Interactive labels attached as children to the circle objects so users can see the site name when clicking on them.
- **MarkerCluster:** A clustering container plugin object used to hold and group multiple individual launch record markers together dynamically.
- **folium.Marker with folium.Icon:** Individual geographic markers generated for every single historical launch record in the dataset, styled dynamically with custom colors matching their outcomes.

Build an Interactive Map with Folium

1. Highlighting Main Sites (folium.Circle & folium.Popup)

The plain numerical latitude and longitude values of the launch sites are not intuitively digestible on their own. Creating circles highlights the precise boundary positions of the launch facilities globally. Attaching a Popup ensures that users can easily click to identify which specific site they are evaluating.

2. Immediate Identification (folium.map.Marker with DivIcon)

Standard popups require a user click to show text. By incorporating custom text-based DivIcon markers directly on top of the coordinates, the launch site names remain visible at a glance without any manual interaction, improving map readability.

3. Visualizing Performance and Handling Density (MarkerCluster & Dynamic folium.Marker)

The dataset includes dozens of individual launch rows, but every single launch occurs from just one of the four main sites.

- If regular markers are plotted for all rows, they would stack directly on top of each other, cluttering the map and hiding data. Using a **MarkerCluster** groups these overlapping nodes elegantly, expanding into individual markers only when a user zooms in.
- The individual markers use conditional color logic mapping directly to launch performance. Successful launches (class=1) are colored **green**, while failed launches (class=0) are colored **red**. This visual encoding allows an analyst to instantly assess which launch sites yield higher success rates versus failure rates.
- GitHub URL interactive map with Folium map: https://github.com/buHtiGDasda/IBM/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

1. Interaction Components

Launch Site Drop-down Menu (`dcc.Dropdown`)

- Allows the user to select either "All Sites" or a specific launch site (e.g., 'CCAFS LC-40', 'VAFB SLC-4E', etc.). It features a text placeholder and text search capabilities.
- It serves as the primary filter for the entire dashboard. Launch conditions, success rates, and optimal payload weights heavily depend on the physical geography and infrastructure of each specific site. This dropdown lets stakeholders easily segment and compare individual launch pad performances.

Payload Range Slider (`dcc.RangeSlider`)

- Provides a dual-handle slider spanning from \$0\$ to \$10,000\text{ kg}\$ with marks every \$2,000\text{ kg}\$. The user can adjust the minimum and maximum boundaries to filter a specific weight range.
- Rocket performance changes dramatically depending on how heavy the payload is. This slider allows to isolate heavy, medium, or light payloads to observe how weight restrictions affect the success or failure of a mission.

Build a Dashboard with Plotly Dash

2. Visualizations & Dynamic Plots

Success Rate Pie Chart (dcc.Graph)

- **What it does:**
 - When "**All Sites**" is selected, it displays the total number of successful launches contributed by each launch site.
 - When a **specific site** is selected, it shifts to display a breakdown of successes (1) versus failures (0) exclusively for that location.
- **Why it was added:** High-level metrics need to answer a simple question instantly: *"Who is performing best?"* When viewing all sites, the pie chart immediately highlights which launch pad has contributed the most to overall program success. When drilling down into a single site, it acts as a quick gauge of that specific site's reliability and operational track record.

Payload vs. Success Correlation Scatter Plot (dcc.Graph)

- **What it does:** Plots the **Payload Mass (kg)** on the x-axis against the **Class** (0 for failure, 1 for success) on the y-axis. Every point is color-coded by its Booster Version Category. The data dynamically filters based on both the selected launch site and the active range slider.
- **Why it was added:** While the pie chart shows *how many* successes occurred, this scatter plot explains *why* or *under what conditions* they occurred. By plotting mass against binary outcomes, users can spot clusters where launches are highly successful (e.g., sweet spots in payload mass). Color-coding by booster version also lets analysts see if specific iterations or upgrades of the Falcon 9 rocket perform better under heavy load profiles than others.

Predictive Analysis (Classification)

Model Development Process

The process of building, evaluating, and finding the best-performing classification model followed a standard data science pipeline. It began with data preparation, moved through multi-model hyperparameter tuning, and concluded with a final evaluation on a test set to determine the best model.

Key Phases of the Process

- **Data Preparation & Preprocessing:** Defined the target labels from the `Class` column, standardized the features using `StandardScaler` to bring them to a common scale, and split the data into training and test sets (80/20 split).
- **Hyperparameter Tuning & Training:** Trained four distinct classification algorithms: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). Used `GridSearchCV` with 10-fold cross-validation (`cv=10`) on the training data to find the optimal hyperparameters for each model.
- **Evaluation & Comparison:** Evaluated each tuned model based on its training cross-validation score and its accuracy score on the unseen test dataset. Confusion matrices were plotted to analyze the breakdown of true positives and false positives.

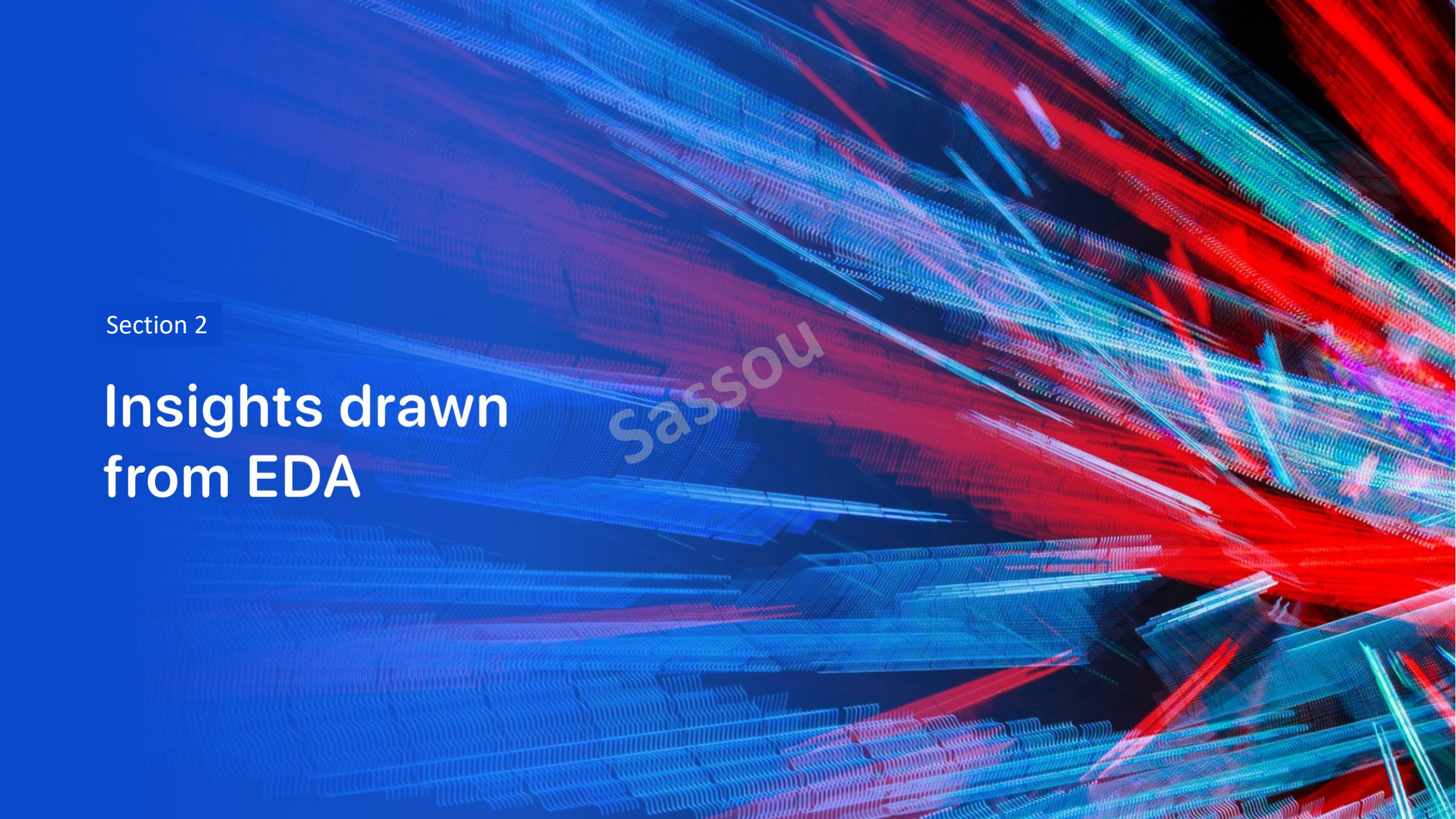
Predictive Analysis (Classification)

- **Model Performance Summary**
- After tuning, the best parameters and test accuracies achieved were as follows:

Model	Best Hyperparameters	Validation Accuracy	Test Accuracy
Logistic Regression	C: 0.01, penalty: 'l2', solver: 'lbfgs'	84.64%	83.33%
SVM		84.82%	83.33%
Decision Tree		88.75%	88.88%
KNN		84.82%	83.33%

Predictive Analysis (Classification)



The background is a dynamic, abstract composition of numerous thin, overlapping lines and streaks in shades of blue, red, and teal. These lines create a sense of motion and depth, resembling a digital or data-driven environment. A large, semi-transparent watermark with the word 'Sassou' is oriented diagonally across the center of the image.

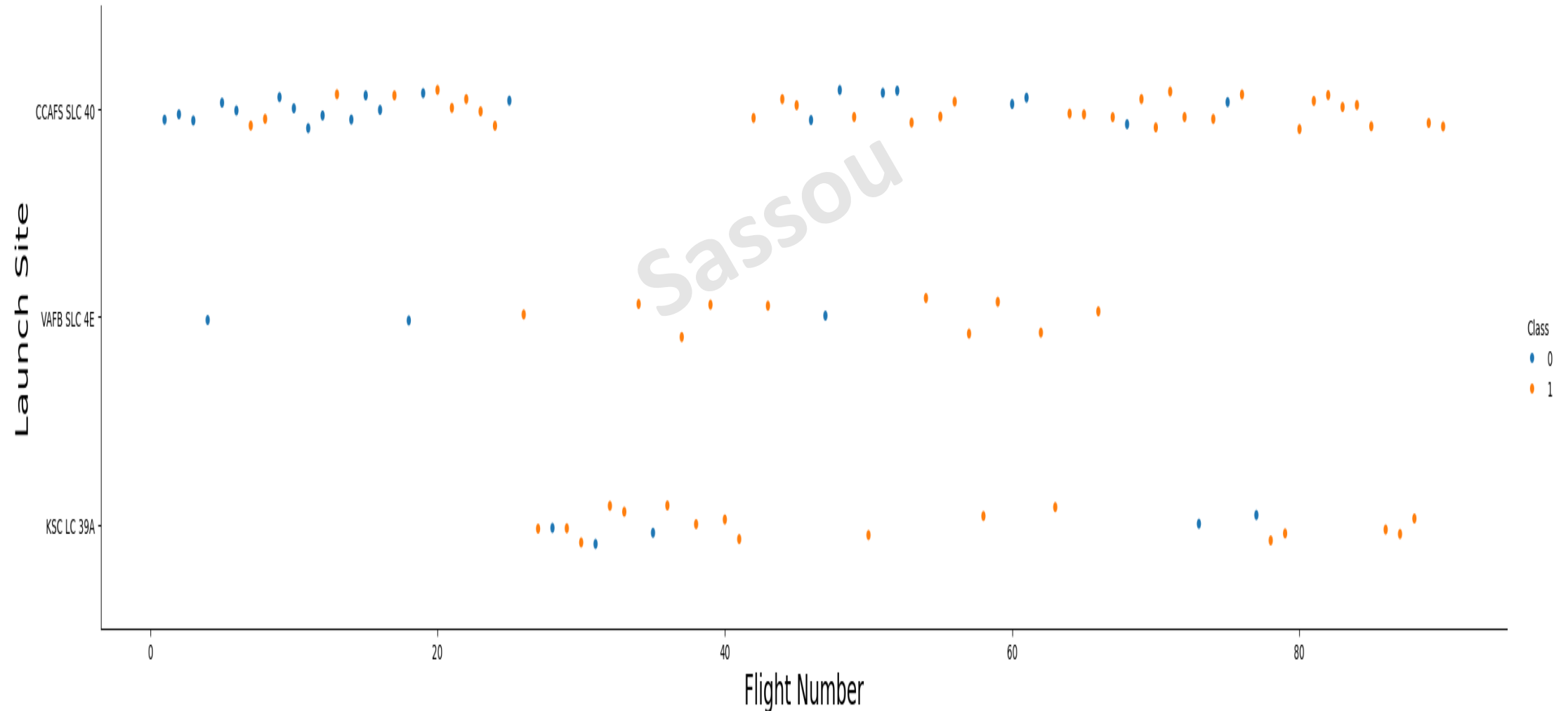
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- **Distribution of Launch Sites:**
- **CCAFS SLC 40** (Cape Canaveral) was the primary site used for early flights (Flight Numbers 1 to 20) and remains heavily utilized throughout the entire timeline.
- **VAFB SLC 4E** (Vandenberg Air Force Base) is used less frequently and typically hosts specific polar orbit or observation satellite missions.
- **KSC LC 39A** (Kennedy Space Center) began seeing operational usage later in the timeline (around Flight Number 20+).
- **Launch Success Trends Over Time:**
- In the early stages of the flight program (lower flight numbers), there is a higher concentration of failures (0, marked in red), particularly at the CCAFS SLC 40 site.
- As the **Flight Number increases**, the frequency of successful landings (1, marked in green) increases significantly across all launch sites. This trend illustrates SpaceX's rapid learning curve, refinement of engineering practices, and mastery of first-stage reusability over consecutive launch attempts.

Here is the scatter plot visualizing the relationship between **Flight Number** and **Launch Site**, with the launch outcome color-coded (where 1 indicates a successful first-stage landing and 0 indicates an unsuccessful attempt):

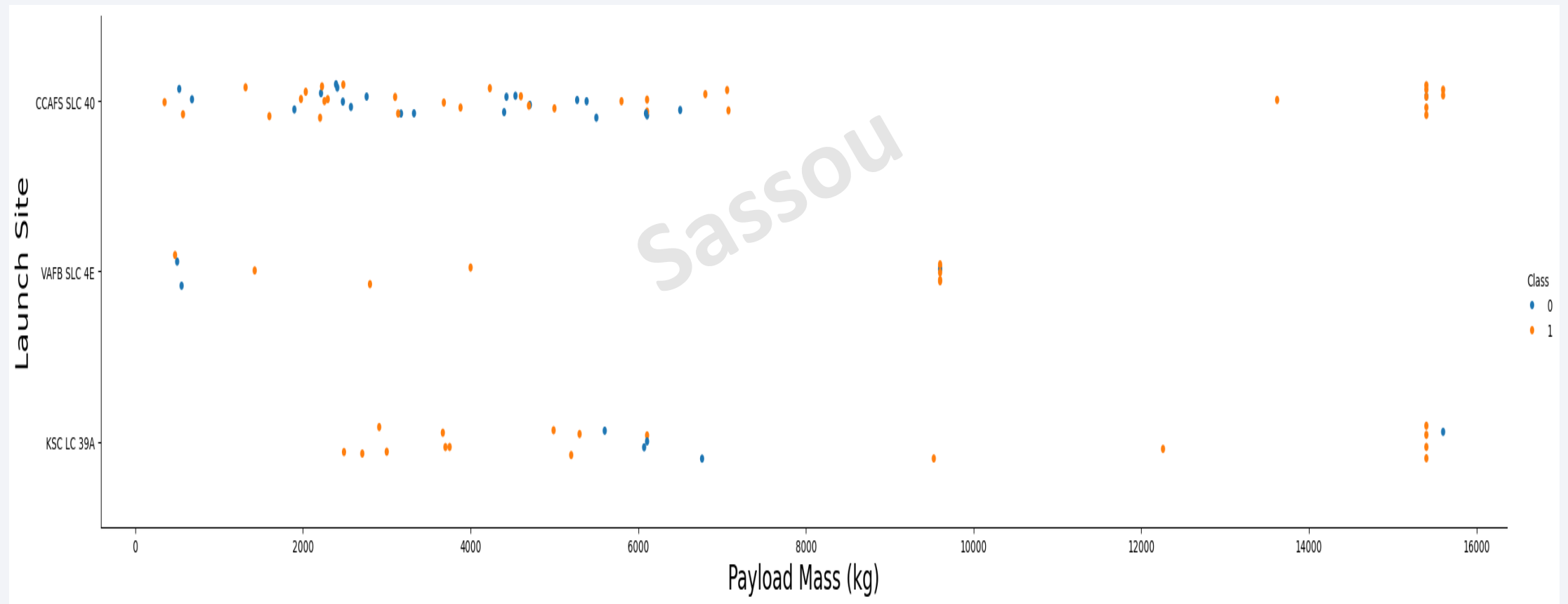


Payload vs. Launch Site

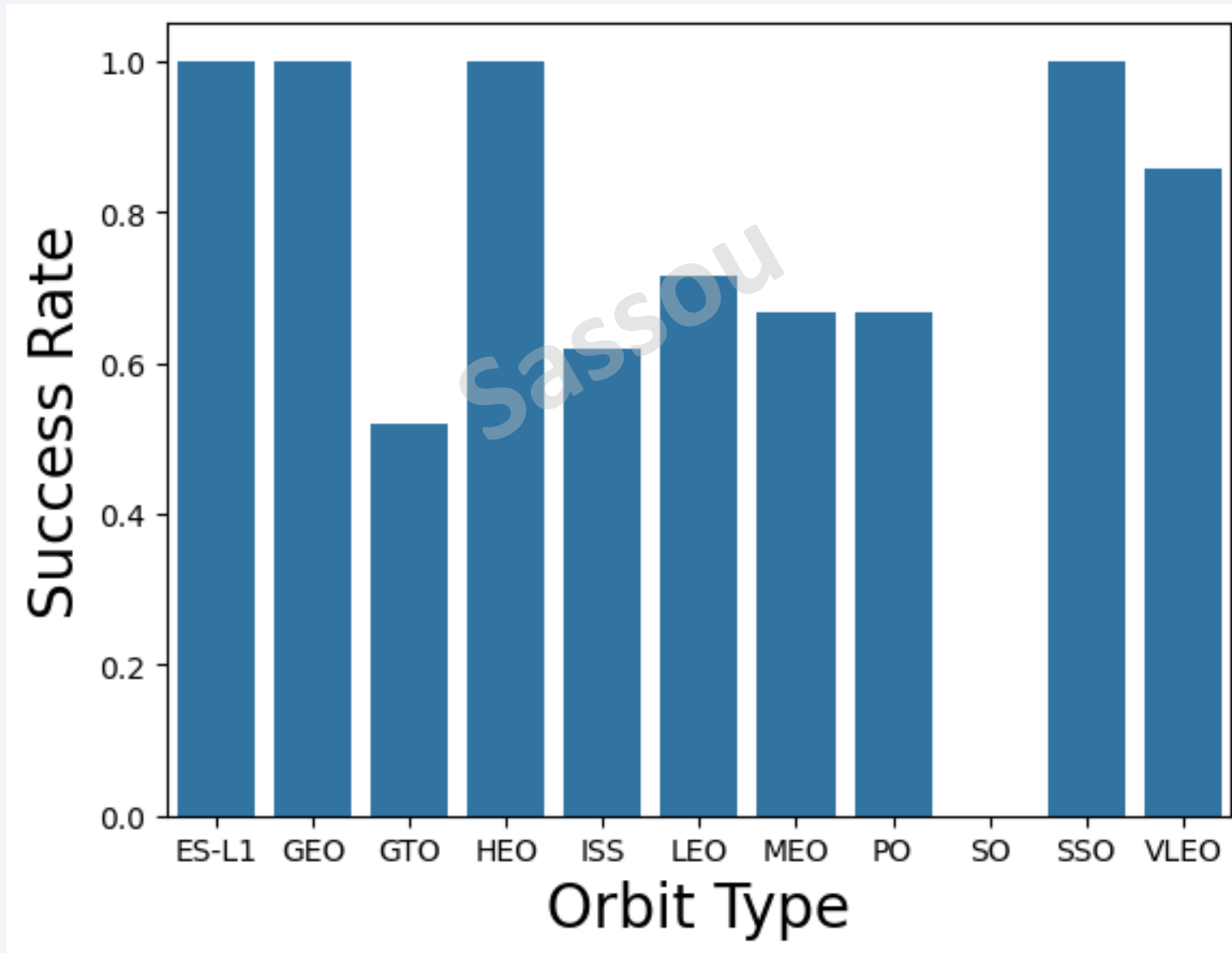
- **Launch Site Distribution by Payload Capacity:**
- **VAFB SLC 4E (Vandenberg Space Force Base):** This site is primary for polar orbits and handles light-to-medium payloads, typically capped below **10,000 kg**.
- **KSC LC 39A (Kennedy Space Center):** This historic site handles medium-to-heavy payloads, with almost no flights carrying payloads under **2,500 kg**.
- **CCAFS SLC 40 (Cape Canaveral Space Force Station):** As SpaceX's most active workhorse pad, it is used for the full spectrum of missions, from very light test payloads and ISS resupply runs to massive heavy commercial satellites over **15,000 kg**.
- **Impact of Payload Mass on Launch Success:**
- For exceptionally heavy payloads (greater than **10,000 kg**), the success rate of the first-stage recovery is remarkably high across both Cape Canaveral (CCAFS SLC 40) and Kennedy Space Center (KSC LC 39A).
- Many of the earlier recorded failures (represented in red) occurred at lower payload masses, which corresponds to earlier, experimental iterations of the Falcon 9 recovery program.

Payload vs. Launch Site

Here is the scatter plot visualizing the relationship between Payload Mass and Launch Site, color-coded by the successful or failed landing outcome of the Falcon 9 first stage.



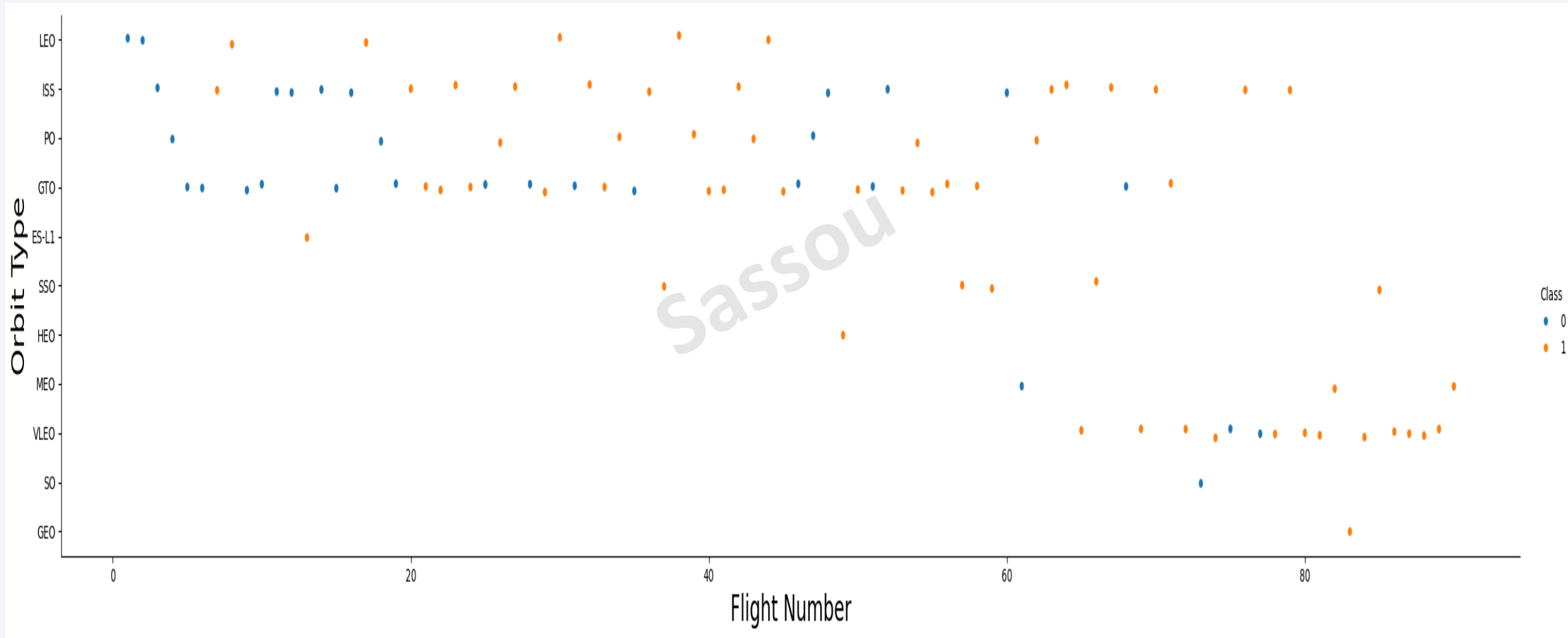
Success Rate vs. Orbit Type



Success Rate vs. Orbit Type

- **Flight Number vs. Success Rate:** As the **Flight Number** increases (moving from left to right on the x-axis), the concentration of successful landings (**Class 1** / orange dots) increases significantly. This indicates that as SpaceX gained more operational experience over subsequent launches, their first-stage recovery capability dramatically improved.
- **Payload Mass Effect:** Higher **Payload Masses** (heavier payloads closer to the top of the y-axis) do not strictly correlate with failure. In fact, even with very massive payloads (e.g., in the 10,000 to 15,000 kg range), the first stage often returns and lands successfully, especially in the later flight numbers.
- **Orbit Dynamics (Inferred):** Different orbits require different launch profiles and payload capacities. For instance, heavy payloads going to lower orbits (like LEO/ISS) show a high density of success in later flights compared to the early test flights where failures (**Class 0** / blue dots) were more common.

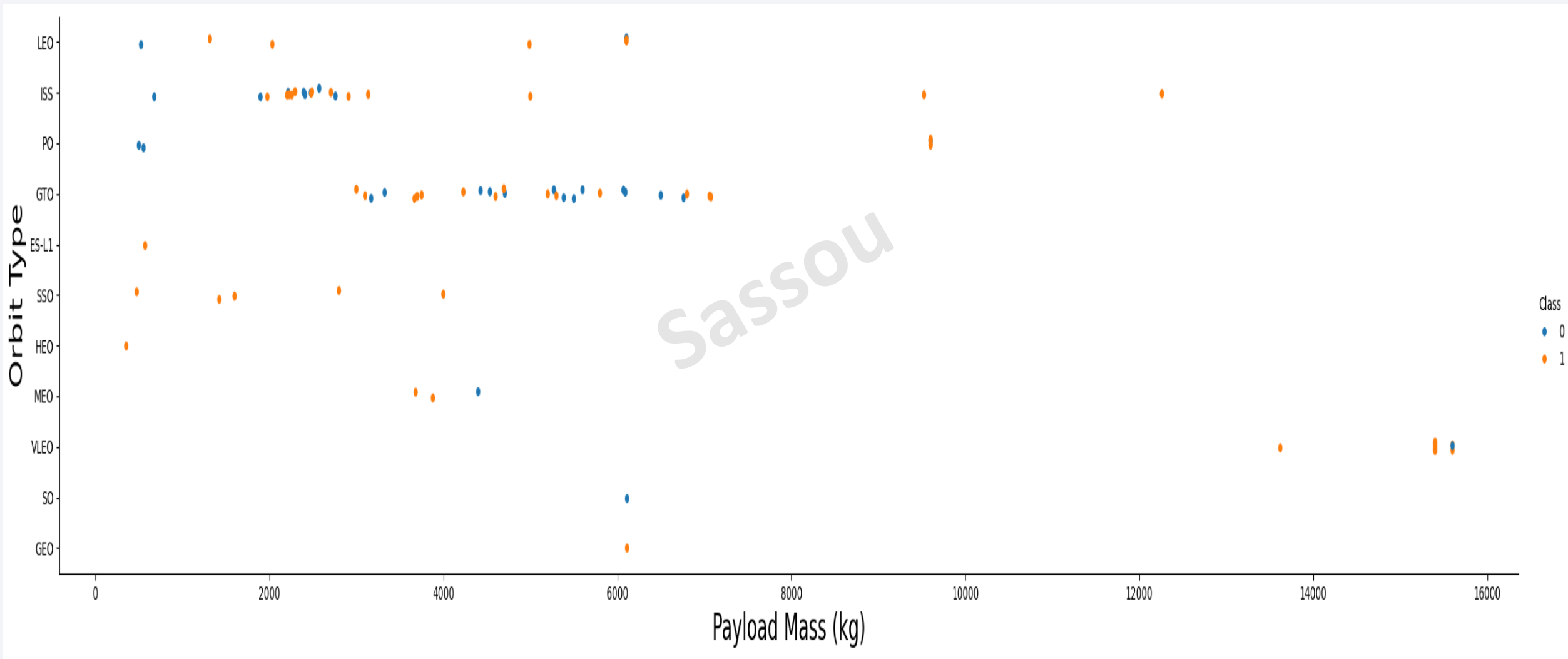
Flight Number vs. Orbit Type



Flight Number vs. Orbit Type

- **Launch Success Trend Over Time:** The color-coding maps the outcome of each flight where **Class 0** represents an unsuccessful landing attempt (blue points) and **Class 1** represents a successful landing (orange points). As the **Flight Number** increases over time, there is a clear and dominant shift from blue points to orange points across almost all orbits, indicating that SpaceX's landing success rate improved drastically with experience.
- **Orbit-Specific Patterns:**
- **GTO (Geostationary Transfer Orbit):** Shows a massive concentration of flights. Early GTO flights struggled with landing success (more blue points), but later flights heavily favor successful recoveries (orange points).
- **ISS (International Space Station) & LEO (Low Earth Orbit):** These orbits show a very high success rate for first-stage landings overall, especially as the flight numbers progressed.
- **VLEO (Very Low Earth Orbit):** Typically associated with later flight numbers (the right side of the graph), these show nearly perfect clusters of successful landings.
- **SO, GEO, and HEO:** These display very sparse data points, indicating rarer or specialized mission profiles.

Payload vs. Orbit Type



Payload vs. Orbit Type

- **Distinct Payload Thresholds by Orbit:**
- **Geostationary Transfer Orbit (GTO):** Displays a tight, high-density cluster of payloads typically ranging between **3,000 kg and 7,000 kg**.
- **International Space Station (ISS):** Features a tightly grouped segment predominantly focused between **2,000 kg and 4,000 kg**.
- **Low Earth Orbit (LEO):** Exhibits highly variable payloads, spanning all the way up to **15,000+ kg** (largely driven by heavy constellations like Starlink).
- **Landing Success Correlation (Class Hue):**
- **Orange Dots (Class=1):** Signify a **successful** first-stage recovery.
- **Blue Dots (Class=0):** Signify an **unsuccessful** or unrecovered landing attempt.
- As flight iterations progressed, the proportion of orange dots increased dramatically across all target orbits, illustrating SpaceX's evolution and maturity in precision propulsion landings.
- **Heavy Payload Resiliency:** Even at extreme payload limits (>15,000 kg) within LEO configurations, the first stage demonstrates a near-flawless successful landing record. This indicates that high-mass profile launches do not inherently preclude recovery when the structural energy margins match low-altitude orbit designs.

Launch Success Yearly Trend



Launch Success Yearly Trend

- **X-Axis (Flight Number):** Represents the continuous chronological progression of SpaceX launch attempts.
- **Y-Axis (Payload Mass in kg):** Details the weight carried by the Falcon 9 rocket for each mission.
- **Hue/Color Coding (Class):** Represents the landing outcome (e.g., 0 for an unsuccessful landing iteration, 1 for a successful first-stage recovery).
- **Insights Drawn:**
 - **Experience Factor:** As the Flight Number grows larger over time, the frequency of successful landing events (Class 1) increases significantly, showing operational improvements.
 - **Payload Resilience:** Even at higher payload volumes (surpassing 6,000 to 10,000 kg), the system demonstrates highly reliable retrieval patterns later in the series.

All Launch Site Names

- The unique launch sites used for the Falcon 9 first-stage landing predictions are:
- **CCAFS SLC 40** (Cape Canaveral Air Force Station Space Launch Complex 40)
- **VAFB SLC 4E** (Vandenberg Air Force Base Space Launch Complex 4E)
- **KSC LC 39A** (Kennedy Space Center Launch Complex 39A)
- **The Method:** The `.unique()` function in Pandas scans the LaunchSite column of the DataFrame (df) and eliminates duplicate values, returning an array of only the distinct operational locations.
- **The Sites:** These three locations represent SpaceX's main coastal launch sites in the United States, covering both the East Coast (Florida) and West Coast (California) paths to satisfy different orbit trajectories (such as LEO, GTO, and polar orbits).

Launch Site Names Begin with 'CCA'

- Here are 5 records from the dataset where the launch site begins with CCA (specifically, CCAFS SLC 40):

Flight Number	Date	Booster Version	Payload Mass (kg)	Orbit	Launch Site	Outcome
1	2010-06-04	Falcon 9	6104.96	LEO	CCAFS SLC 40	None None
2	2012-05-22	Falcon 9	525.00	LEO	CCAFS SLC 40	None None
3	2013-03-01	Falcon 9	677.00	ISS	CCAFS SLC 40	None None
5	2013-12-03	Falcon 9	3170.00	GTO	CCAFS SLC 40	None None
6	2014-01-06	Falcon 9	3325.00	GTO	CCAFS SLC 40	None None

Total Payload Mass

- Total payload carried by boosters from NASA
- **Total Payload Mass: 48,213 kg**
- This calculation uses the SUM() function to aggregate the total weights stored in the payload mass column (PAYLOAD_MASS__KG_). The WHERE clause filters out any other commercial or military clients, isolating only the resupply missions conducted on behalf of **NASA (Commercial Resupply Services)**.

Average Payload Mass by F9 v1.1

- **Average Payload Mass: 2,928.4 kg**
- The query uses the AVG() aggregate function to compute the mean value of the PAYLOAD_MASS__KG_ column. The WHERE clause filters the dataset to target only rows matching the exact booster version string 'F9 v1.1'. The resulting average indicates that missions utilizing the standard Falcon 9 v1.1 configuration typically carried payloads averaging just under 3 metric tons, primarily consisting of early Dragon cargo missions to the ISS and various commercial satellites.

First Successful Ground Landing Date

- Based on the SpaceX Falcon 9 dataset, the date of the first successful landing outcome on a ground pad is **2015-12-22**.
- During the exploratory data analysis and SQL querying phase of the SpaceX dataset, the Outcome column tracking landing results is filtered for rows representing a success on land (often recorded as "Success (ground pad)" or "True RTLS"). Finding the minimum date (MIN(Date)) from this filtered subset isolates the historic milestone launch.

Successful Drone Ship Landing with Payload between 4000 and 6000

- Based on the SpaceX dataset, the booster versions that successfully landed on a drone ship with a payload mass greater than 4000 kg but less than 6000 kg are:
- **F9 FT B1022** (Serial: **B1022**)
- **F9 FT B1026** (Serial: **B1026**)
- **Landing Outcome:** Both boosters achieved a landing outcome of "**Success (drone ship)**" (represented as True ASDS in the filtered dataset_part_2.csv columns).
- **Payload Mass Filter:**
- **F9 FT B1022** carried a payload mass of **4,696 kg** during the JCSAT-14 mission.
- **F9 FT B1026** carried a payload mass of **4,600 kg** during the JCSAT-16 mission. Both of these values fall strictly within the specified range of **> 4000 kg** and **< 6000 kg**.

Total Number of Successful and Failure Mission Outcomes

- Based on the SpaceX Falcon 9 dataset (dataset_part_2.csv), the mission outcomes in the Outcome column are mapped to the binary Class variable, where 1 represents a successful first-stage landing and 0 represents a landing failure.
- **Successful Outcomes (Class = 1): 60 launches**
 - True ASDS (Autonomous Spaceport Drone Ship): 41
 - True RTLS (Return To Launch Site): 14
 - True Ocean (Controlled ocean landing): 5
- **Failure Outcomes (Class = 0): 30 launches**
 - None None (No landing attempt/uncontrolled): 19
 - False ASDS (Failed drone ship landing): 6
 - False Ocean (Unsuccessful ocean landing): 2
 - None ASDS (Unsuccessful drone ship attempt): 2
 - False RTLS (Failed return to launch site landing): 1

Out of the **90 total Falcon 9 test flights** recorded in this dataset, SpaceX achieved a successful first-stage landing recovery in **66.67%** of their missions (60 out of 90). The remaining 30 flights resulted in various types of failures, including cases where the stage was intentionally left unrecovered (None None), or where a landing was attempted on a drone ship or ground pad but failed (False or None outcomes).

Boosters Carried Maximum Payload

- **B1048**
- **B1049**
- **B1051**
- **B1056**
- **B1059**
- **B1060**
- **B1058**
- Based on the SpaceX dataset utilized in the assignment, the maximum payload mass carried by any booster is **15,600 kg**

2015 Launch Records

In the dataset, an "**ASDS**" (Autonomous Spaceport Drone Ship) outcome represents a drone ship landing attempt. A prefix of "**False**" explicitly indicates an unsuccessful or failed attempt. During the year 2015, SpaceX made two separate attempts to land the Falcon 9 first-stage booster on a drone ship. Both of these missions were launched from Cape Canaveral (**CCAFS SLC 40**) and both resulted in a failed landing outcome (False ASDS).

Landing Outcome	Booster Version	Launch Site	Date
False ASDS	Falcon 9	CCAFS SLC 40	2015-01-10
False ASDS	Falcon 9	CCAFS SLC 40	2015-04-14

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

During this early operational timeframe (from June 2010 to March 2017), the most frequent landing outcome was **No attempt** with 10 occurrences, reflecting the initial phase of the Falcon 9 program before first-stage recovery became standard practice.

As SpaceX began testing propulsive recovery methods, drone ship attempts yielded an even split of **5 successes** and **5 failures**. Landings on a ground pad (**Success (ground pad)**) were achieved 3 times, while early experimental parachute and uncontrolled ocean tests made up the rest of the lower ranks.

Landing Outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. A large, semi-transparent watermark reading "Sassou" is diagonally across the center of the image.

Section 3

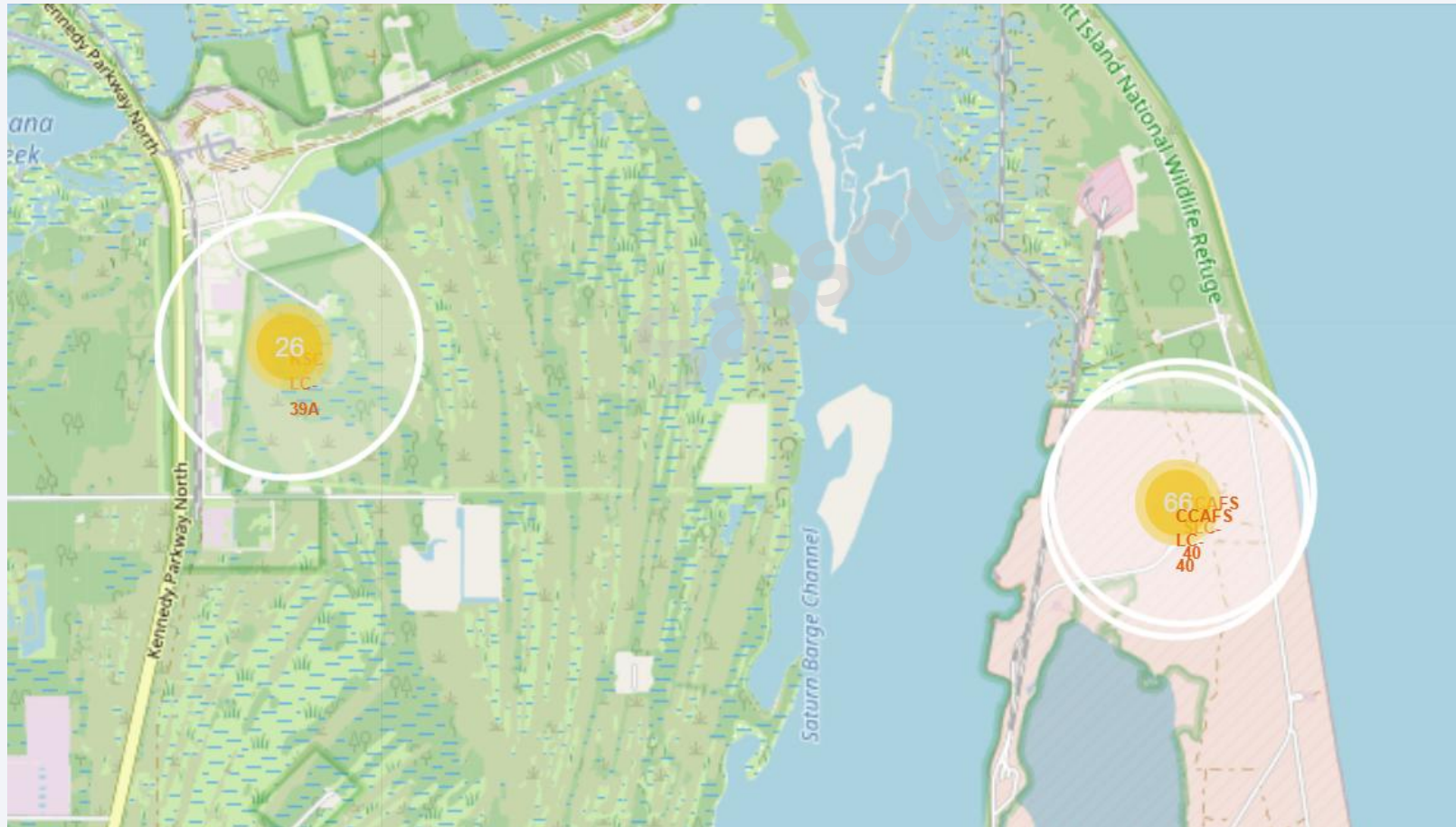
Launch Sites Proximities Analysis

Global Map of SpaceX Launch Site Locations



The sites are clustered in Florida and California, between 28° and 35°N (close to the equator to take advantage of the Earth's rotation during eastward launches, but not exactly on it), and all are located on the coast (for safety regarding rocket debris).

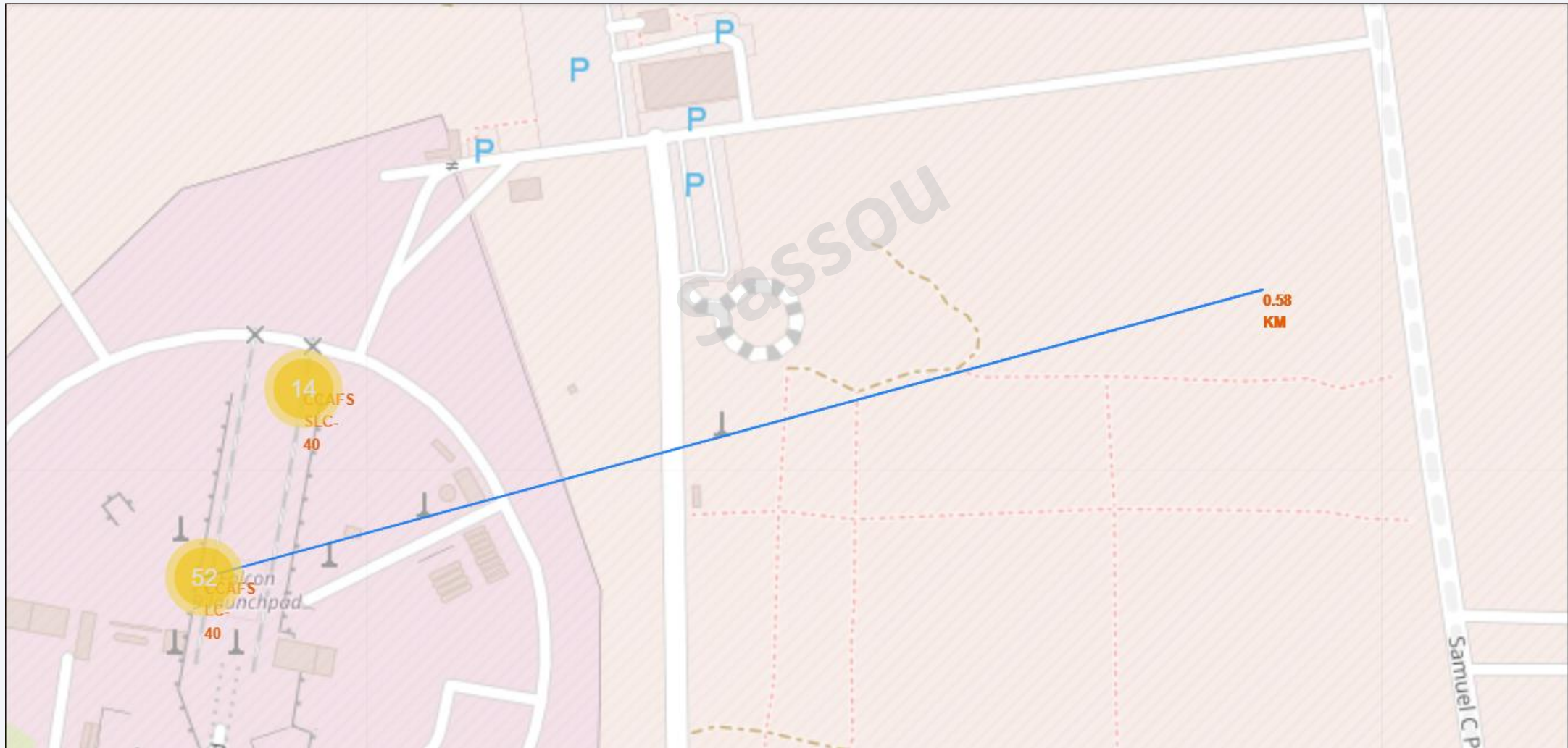
Color-Labeled Launch Outcomes by Launch Site



Color-Labeled Launch Outcomes by Launch Site



Proximity Analysis: CCAFS LC-40 Launch Site to Coastline, Railway, and Highway



Proximity Analysis: CCAFS LC-40 Launch Site to Coastline, Railway, and Highway

- **Coastline** (28.56367, -80.57163) → **≈0.58 km** from the pad — the launch site sits almost directly on the coast, consistent with rockets launching out over open water for safety.
- **Highway** (28.56335, -80.57085) → **≈0.65 km** — a paved road runs close to the site for logistics access.
- **Railway** (28.57206, -80.58525) → **≈1.33 km** — a rail line, likely used for heavy equipment/component transport, is slightly farther but still nearby.
- **City** (Titusville, FL at 28.61208, -80.80757, cell 73) → **≈23.16 km** — the nearest populated center is kept well clear of the pad.
- **Interpretation:** the pattern of short distances to coastline/highway/railway but a long distance to the city supports the general siting logic for launch pads — close enough to transport infrastructure and open water (for downrange safety and barge/road delivery of rocket stages) but far enough from populated areas to limit risk in case of a launch failure.

Section 4

Build a Dashboard with Plotly Dash

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Launch success count for all sites

SpaceX Launch Records Dashboard

All Sites

X v

Total Success Launches By Site



The pie chart illustrates the percentage distribution of total successful launches across the four sites:

KSC LC-39A accounts for the largest share of successful launches at 41.7%.

CCAFS LC-40 follows as the second most active site with 29.2%.

VAFB SLC-4E represents 16.7% of successful launches.

CCAFS SLC-40 holds the smallest share among those listed, at 12.5%.

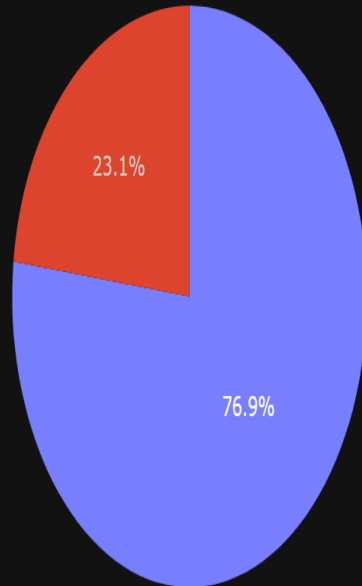
Launch site with highest launch success ratio

SpaceX Launch Records Dashboard

KSC LC-39A

X v

Total Success Launches for Site KSC LC-39A



The data indicates that at the KSC LC-39A launch site:

76.9% of the launches have been successful.

23.1% of the launches have not been successful.

Scatter Plot Analysis: Correlation Between Rocket Payload Mass and Launch Success



Scatter Plot Analysis: Correlation Between Rocket Payload Mass and Launch Success

- **Overall Trends:** Successes (class 1) are distributed across a wide range of payload masses, from approximately 500 kg up to over 9,000 kg. Failures (class 0) are also present throughout the lower and middle payload ranges.
- **Booster Version Performance:**
- **FT (Full Thrust):** This version is heavily represented in the successful launches (class 1) across a broad spectrum of payload weights, ranging from roughly 500 kg to 5,500 kg. It shows a strong overall success rate compared to others.
- **v1.1:** This version appears frequently in the failure category (class 0) across various payload masses, suggesting a lower success rate compared to the FT booster.
- **v1.0, B4, and B5:** These categories have fewer data points overall. Notably, B4 is associated with both successes and failures at higher payload masses, while B5 is represented at a mid-range payload mass as a success.
- **Payload Range:** There is no single "best" payload range for success; rather, the data shows that successful launches occur consistently across the entire spectrum shown, even as payload mass increases toward the 10,000 kg mark.

Scatter Plot Analysis: Correlation Between Rocket Payload Mass and Launch Success



Scatter Plot Analysis: Correlation Between Rocket Payload Mass and Launch Success

- **Performance Evolution:** Newer booster versions (like **B4** and **B5**) demonstrate a higher frequency of success (Class 1) across various payload ranges compared to the older **v1.1** booster, which appears predominantly in the failure category (Class 0) at lower payload masses.
- **Payload and Success:** While successful launches (Class 1) occur across a wide range of payload masses, there is a visible trend where higher payload missions (especially those exceeding 5,000 kg) are predominantly handled by the **FT** and **B4/B5** boosters.
- **Booster Success Patterns:**
 - **FT (Red):** Shows a significant number of successes across a broad payload spectrum (roughly 2,000 kg to 6,500 kg), though it also has several failures in that same range.
 - **B4 (Dark Green):** Displays a very high success rate for heavier payloads, with successful landings recorded even at the high end of the current scale (near 9,000+ kg).
 - **v1.1 (Light Blue):** This version is clustered in the lower payload range (typically under 4,500 kg) and shows a higher density of Class 0 (failures).

Scatter Plot Analysis: Correlation Between Rocket Payload Mass and Launch Success



Scatter Plot Analysis: Correlation Between Rocket Payload Mass and Launch Success

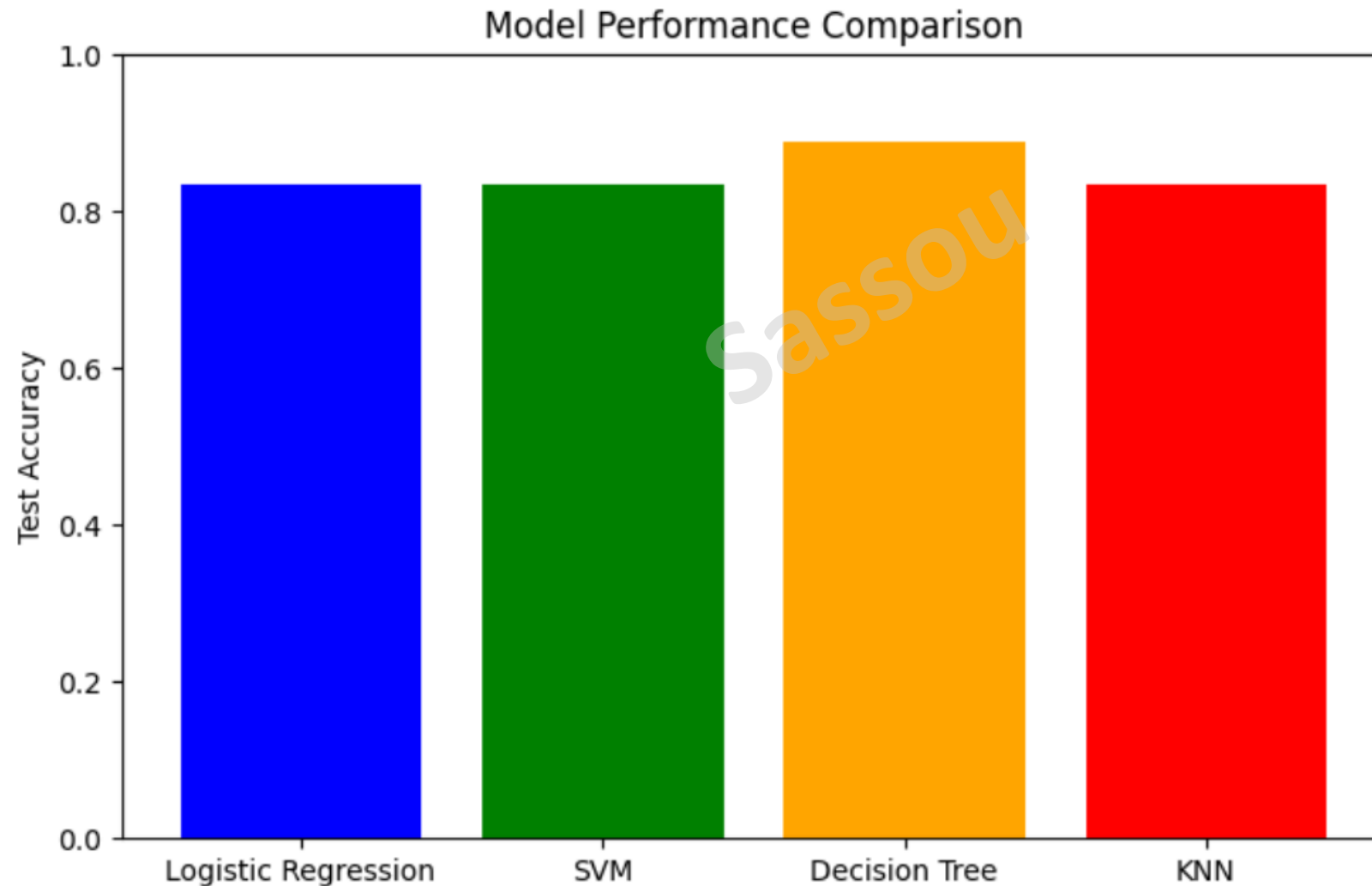
- **General Success Trends:** There is no single "payload range" that guarantees success, as both failures (0) and successes (1) are scattered across a wide range of masses (roughly 2,000 kg to 6,500 kg).
- **Performance by Booster Version:**
- **FT (Red):** This version is heavily represented across the entire payload spectrum shown (approx. 2,000 kg to 7,000 kg) and shows a mix of both successes and failures.
- **B4 (Dark Green):** These boosters appear in both the lower and higher payload ranges. They show successful outcomes even at high payload masses (near 9,500 kg), where other versions are less frequently represented in the plot.
- **v1.1 (Light Blue):** This version appears primarily in the lower payload range (under 5,000 kg).
- **B5 (Purple):** There is very limited data for this version in the current view, making it difficult to draw a broad conclusion about its specific success rate compared to the others.
- **Observation on Payload Mass:** While there are more failures clustered under 6,000 kg, this may simply be due to a higher frequency of launches in that range. Successes are documented throughout the range, suggesting that payload mass alone is not the sole determinant of mission success.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

The best performing method on the test data is: Decision Tree with an accuracy of 0.8888888888888888



Confusion Matrix



Confusion Matrix

- **Matrix Breakdown**
- **True Negatives (Top-Left, 5):** The model correctly predicted that the event "did not land" would not land.
- **False Positives (Top-Right, 1):** The model incorrectly predicted that the event would "land" when it actually "did not land."
- **False Negatives (Bottom-Left, 1):** The model incorrectly predicted that the event "did not land" when it actually "landed."
- **True Positives (Bottom-Right, 11):** The model correctly predicted that the event would "land" and it did land.
- **Performance Summary**
- **Total Correct Predictions:** $5+11=16$
- **Total Incorrect Predictions:** $1+1=2$
- **Accuracy:** The model correctly predicted the outcome in **88.9%** (16/18) of the cases.
- In summary, the model shows a strong performance with 16 correct classifications out of 18 total instances, demonstrating a higher sensitivity and precision for the "landed" category compared to the "did not land" category.

Conclusions

- Falcon 9 first-stage landing success increased steadily with flight number, reflecting SpaceX's engineering maturity
- CCAFS SLC-40 was the most heavily used launch site, handling the full range of payload masses
- Heavier payloads did not reduce landing success — recovery remained reliable even above 10,000–15,000 kg
- The Decision Tree classifier was the best-performing model, achieving 88.9% test accuracy (16/18 correct predictions)
- These results show that first-stage landing outcome can be predicted with good accuracy from launch parameters, supporting cost estimation for competing launch providers
- Given the 88.9% prediction accuracy, a competing launch provider could reliably estimate first-stage recovery — and therefore launch cost — before committing to a mission, directly addressing the cost-competitiveness question raised in the Introduction

Thank you!

Sassou

